



**Subject Name: Programming for Problem Solving**  
**Subject Code: BE01R00121**

|        | <div>CHAPTER NO- 2: FUNDAMENTALS OF C</div> <div>[Solution of MSE-1 Syllabus Topics]</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                            |   |                                          |   |                                                  |   |                                              |   |                                        |   |                                     |   |                                                |   |                                          |   |                                         |   |                                        |    |                                                                                      |   |
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|        | <div>TOPIC:1 Features of C Language, Structure of C Program, Comments, Header Files</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |                            |   |                                          |   |                                                  |   |                                              |   |                                        |   |                                     |   |                                                |   |                                          |   |                                         |   |                                        |    |                                                                                      |   |
|        | <div>DESCRIPTIVE QUESTIONS</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |                            |   |                                          |   |                                                  |   |                                              |   |                                        |   |                                     |   |                                                |   |                                          |   |                                         |   |                                        |    |                                                                                      |   |
| 1      | <div>List the header files used in C Programing with their properties.(Dec-2015-OLD) [NLJIET]</div> <table><tr><th>Sr.No.</th><th>Header Files &amp; Description</th></tr><tr><td>1</td><td><b>stdio.h</b><br/>Input/Output functions</td></tr><tr><td>2</td><td><b>conio.h</b><br/>Console Input/Output functions</td></tr><tr><td>3</td><td><b>stdlib.h</b><br/>General utility functions</td></tr><tr><td>4</td><td><b>math.h</b><br/>Mathematics functions</td></tr><tr><td>5</td><td><b>string.h</b><br/>String functions</td></tr><tr><td>6</td><td><b>ctype.h</b><br/>Character handling functions</td></tr><tr><td>7</td><td><b>time.h</b><br/>Date and time functions</td></tr><tr><td>8</td><td><b>float.h</b><br/>Limits of float types</td></tr><tr><td>9</td><td><b>limits.h</b><br/>Size of basic types</td></tr><tr><td>10</td><td><b>wctype.h</b><br/>Functions to determine the type contained in wide character data.</td></tr></table> | Sr.No. | Header Files & Description | 1 | <b>stdio.h</b><br>Input/Output functions | 2 | <b>conio.h</b><br>Console Input/Output functions | 3 | <b>stdlib.h</b><br>General utility functions | 4 | <b>math.h</b><br>Mathematics functions | 5 | <b>string.h</b><br>String functions | 6 | <b>ctype.h</b><br>Character handling functions | 7 | <b>time.h</b><br>Date and time functions | 8 | <b>float.h</b><br>Limits of float types | 9 | <b>limits.h</b><br>Size of basic types | 10 | <b>wctype.h</b><br>Functions to determine the type contained in wide character data. | 4 |
| Sr.No. | Header Files & Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                            |   |                                          |   |                                                  |   |                                              |   |                                        |   |                                     |   |                                                |   |                                          |   |                                         |   |                                        |    |                                                                                      |   |
| 1      | <b>stdio.h</b><br>Input/Output functions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |                            |   |                                          |   |                                                  |   |                                              |   |                                        |   |                                     |   |                                                |   |                                          |   |                                         |   |                                        |    |                                                                                      |   |
| 2      | <b>conio.h</b><br>Console Input/Output functions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |                            |   |                                          |   |                                                  |   |                                              |   |                                        |   |                                     |   |                                                |   |                                          |   |                                         |   |                                        |    |                                                                                      |   |
| 3      | <b>stdlib.h</b><br>General utility functions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |                            |   |                                          |   |                                                  |   |                                              |   |                                        |   |                                     |   |                                                |   |                                          |   |                                         |   |                                        |    |                                                                                      |   |
| 4      | <b>math.h</b><br>Mathematics functions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |                            |   |                                          |   |                                                  |   |                                              |   |                                        |   |                                     |   |                                                |   |                                          |   |                                         |   |                                        |    |                                                                                      |   |
| 5      | <b>string.h</b><br>String functions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |                            |   |                                          |   |                                                  |   |                                              |   |                                        |   |                                     |   |                                                |   |                                          |   |                                         |   |                                        |    |                                                                                      |   |
| 6      | <b>ctype.h</b><br>Character handling functions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |                            |   |                                          |   |                                                  |   |                                              |   |                                        |   |                                     |   |                                                |   |                                          |   |                                         |   |                                        |    |                                                                                      |   |
| 7      | <b>time.h</b><br>Date and time functions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |                            |   |                                          |   |                                                  |   |                                              |   |                                        |   |                                     |   |                                                |   |                                          |   |                                         |   |                                        |    |                                                                                      |   |
| 8      | <b>float.h</b><br>Limits of float types                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |                            |   |                                          |   |                                                  |   |                                              |   |                                        |   |                                     |   |                                                |   |                                          |   |                                         |   |                                        |    |                                                                                      |   |
| 9      | <b>limits.h</b><br>Size of basic types                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |                            |   |                                          |   |                                                  |   |                                              |   |                                        |   |                                     |   |                                                |   |                                          |   |                                         |   |                                        |    |                                                                                      |   |
| 10     | <b>wctype.h</b><br>Functions to determine the type contained in wide character data.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |                            |   |                                          |   |                                                  |   |                                              |   |                                        |   |                                     |   |                                                |   |                                          |   |                                         |   |                                        |    |                                                                                      |   |
|        | <div><ul style="list-style-type: none"><li>• In C language, header files contain the set of predefined standard library functions. The “#include” pre-processing directive is used to include the header files with “.h” extension in the program.</li><li>• Types of Header Files in C: There are 2 types of header files such as:-</li><li>• <b>Standard Library Header Files:</b> - These are pre-existing header files which are available in C.</li><li>• <b>User-Defined Header Files:</b> - The header files which are defined by the user then they are called user-defined header files. The #define directive is used to define a header file.</li><li>• <b>Syntax of Header File in C</b><br/>Way to include a header file in your program:-</li></ul></div>                                                                                                                                                                                   |        |                            |   |                                          |   |                                                  |   |                                              |   |                                        |   |                                     |   |                                                |   |                                          |   |                                         |   |                                        |    |                                                                                      |   |



|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                 |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
|   | <ul style="list-style-type: none"> <li>• <code>#include&lt;headerFilename&gt;</code></li> <li>• The <code>stdio.h</code> header file allows us to perform input and output operations in C. The functions like <code>printf()</code> and <code>scanf()</code> are used to display output and take input from a user, respectively. This library allows us to communicate with users easily.</li> <li>• <code>conio.h</code> is a header file in which there are many built-in functions embedded in it they generally perform <b>input/output</b> on the console i.e., it is used to take input from the keyboard given by the user and display output on the screen. Some functions of <code>conio.h</code> such as <code>getch()</code> is used to hold the screen till the user has pressed any key.</li> <li>• <code>Math.h</code> is a header file which supports all mathematical functions like <code>pow()</code>, <code>sqrt()</code> to find power and square root respectively.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                 |
| 2 | <p><b>Discuss the important of <code>stdio.h</code> and <code>conio.h</code> header file. (Jan-2019-OLD) [NLJIET]</b></p> <p><b>Discuss the important of <code>stdio.h</code> header file. (Jun-2016-OLD) [NLJIET]</b></p> <p><b>Show the important of <code>stdio.h</code> header file. (Jan-2020-NEW) [NLJIET]</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p><b>4</b></p> <p><b>3</b></p> <p><b>3</b></p> |
|   | <ul style="list-style-type: none"> <li>• <b>STDIO</b> means Standard Input Output. It has some pre-defined functions like <code>int printf()</code>, <code>scanf()</code> etc.</li> <li>• <b>CONIO</b> stands for Console Input Output which has some functions like <code>clrscr()</code>, <code>getch()</code> etc.</li> <li>• <code>#include</code> is a preprocessor directory.</li> <li>• It will include the file which is given within the angle brackets "&lt;&gt;" into the current source file.<br/><b>Example.</b></li> <li>• The <code>stdio.h</code> header file allows us to perform input and output operations in C. The functions like <code>printf()</code> and <code>scanf()</code> are used to display output and take input from a user, respectively. This library allows us to communicate with users easily.</li> <li>• <code>conio.h</code> is a header file in which there are many built-in functions embedded in it they generally perform <b>input/output</b> on the console i.e., it is used to take input from the keyboard given by the user and display output on the screen. Some functions of <code>conio.h</code> such as <code>getch()</code> is used to hold the screen till the user has pressed any key.</li> <li>• <code>Math.h</code> is a header file which supports all mathematical functions like <code>pow()</code>, <code>sqrt()</code> to find power and square root respectively.</li> </ul> |                                                 |
| 3 | <p><b>What are header files? Name at least 3 with its usage. (Jun-2019-NEW) [NLJIET]</b></p> <p><b>List any three header file with its usage. (Nov-2020-NEW)(Aug-2023-</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p><b>3</b></p> <p><b>3</b></p>                 |

**NEW)[NLJIET]**

A header file is a file with extension **.h** which contains C function declarations and macro definitions to be shared between several source files. There are two types of header files: the files that the programmer writes and the files that comes with your compiler.

- **STDIO** means Standard Input Output. It has some pre-defined functions like `int printf()`, `scanf()` etc. **CONIO** stands for Console Input Output which has some functions like `clrscr()`, `getch()` etc.
- `#include` is a preprocessor directory.
- It will include the file which is given within the angle brackets "<>" into the current source file.
- If we want to use `printf` or `scanf` function in our program, we should include the `stdio.h` header file in our source code. Otherwise, our program doesn't know what is the definition of `printf` or `scanf` and it will throw error/warning saying that implicit declaration of built-in function 'printf'

| Name of the header file | Description of the header file                  | Functions available in the header file                                                                                                                                                          |
|-------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>alloc.h</code>    | Memory management functions                     | <code>calloc()</code> , <code>malloc()</code> , <code>free()</code> , <code>realloc()</code> etc..                                                                                              |
| <code>complex.h</code>  | Complex math functions                          | <code>asin()</code> , <code>atan()</code> , <code>arg()</code> , <code>tanh()</code> etc..                                                                                                      |
| <code>ctype.h</code>    | Character oriented functions                    | <code>toupper()</code> , <code>tolower()</code> , <code>islower()</code> , <code>isupper()</code> etc..                                                                                         |
| <code>dos.h</code>      | DOS functions                                   | <code>getdate()</code> , <code>gettime()</code> , <code>sleep()</code> etc..                                                                                                                    |
| <code>graphics.h</code> | Graphics related functions                      | <code>arc()</code> , <code>bar()</code> , <code>circle()</code> , <code>getx()</code> etc..                                                                                                     |
| <code>process.h</code>  | Process related functions                       | <code>exit()</code> , <code>abort()</code> , <code>system()</code> etc..                                                                                                                        |
| <code>stdio.h</code>    | Standard I/O functions                          | <code>puts()</code> , <code>gets()</code> , <code>fopen()</code> , <code>fclose()</code> , <code>printf()</code> , <code>scanf()</code> , <code>getchar()</code> , <code>putchar()</code> etc.. |
| <code>stdlib.h</code>   | Standard library functions                      | <code>atoi()</code> , <code>time()</code> , <code>abort()</code> etc..                                                                                                                          |
| <code>string.h</code>   | String manipulation functions                   | <code>strcpy()</code> , <code>strcat()</code> , <code>strlen()</code> , etc..                                                                                                                   |
| <code>time.h</code>     | Declare functions that manipulate time and date | <code>time()</code> , <code>stime()</code> , <code>diffime()</code> , <code>localtime()</code> etc..                                                                                            |

- The `stdio.h` header file allows us to perform input and output operations in C. The functions like `printf()` and `scanf()` are used to display output and take input from a user, respectively. This library allows us to communicate with users easily.
- `conio.h` is a header file in which there are many built-in functions embedded in it they generally perform **input/output** on the console i.e., it is used to take input from the keyboard given by the user and display output on the screen. Some functions of



|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|   | <p>conio.h such as getch() is used to hold the screen till the user has pressed any key.</p> <ul style="list-style-type: none"> <li>Math.h is a header file which supports all mathematical functions like pow(), sqrt() to find power and square root respectively.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |
| 4 | <b>Explain following pre-processor directives with example.(Mar-2023-OLD)[NLJIET]</b><br><b>1. #define</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 4 |
|   | <p>A preprocessor directive is usually placed in the top of the source code in a separate line beginning with the character "#", followed by directive name and an optional white space before and after it.</p> <p>To execute a preprocessor program on a certain statement, some of the preprocessor directives types are:</p> <ul style="list-style-type: none"> <li>#define: It substitutes a preprocessor using macro.</li> <li>#include: It helps to insert a certain header from another file.</li> <li>#undef: It undefines a certain preprocessor macro.</li> <li>#ifdef: It returns true if a certain macro is defined.</li> <li>#ifndef: It returns true if a certain macro is not defined.</li> </ul> <p>#if, #elif, #else, and #endif: It tests the program using a certain condition; these directives can be nested too.</p> <p><b>C #define</b></p> <p>The #define preprocessor directive is used to define constant or micro substitution. It can use any basic data type.</p> <p>Syntax:</p> <p>#define token value</p> <p>an example of #define to define a constant.</p> <pre>#include &lt;stdio.h&gt;  #define PI 3.14  main() {  printf("%f",PI);  }</pre> |   |

Output:

3.140000

**5 Explain C tokens in detail(Jan-2024-NEW)[NLJIET]****7**

A token in C can be defined as the smallest individual element of the C programming language that is meaningful to the compiler. It is the basic component of a C program.

**Types of Tokens in C**

The tokens of C language can be classified into six types based on the functions they are used to perform. The types of C tokens are as follows:

**Keywords****Identifiers****Constants****Strings****Special Symbols****Operators****1. C Token – Keywords**

The [keywords](#) are pre-defined or reserved words in a programming language. Each keyword is meant to perform a specific function in a program. Since keywords are referred names for a compiler, they can't be used as variable names because by doing so, we are trying to assign a new meaning to the keyword which is not allowed. You cannot redefine keywords. However, you can specify the text to be substituted for keywords before compilation by using C preprocessor directives. C language supports **32** keywords which are given below:

|              |               |                 |                |
|--------------|---------------|-----------------|----------------|
| <b>auto</b>  | <b>double</b> | <b>int</b>      | <b>struct</b>  |
| <b>break</b> | <b>else</b>   | <b>long</b>     | <b>switch</b>  |
| <b>case</b>  | <b>enum</b>   | <b>register</b> | <b>typedef</b> |



|                 |               |               |                 |
|-----------------|---------------|---------------|-----------------|
| <b>char</b>     | <b>extern</b> | <b>return</b> | <b>union</b>    |
| <b>const</b>    | <b>float</b>  | <b>short</b>  | <b>unsigned</b> |
| <b>continue</b> | <b>for</b>    | <b>signed</b> | <b>void</b>     |
| <b>default</b>  | <b>goto</b>   | <b>sizeof</b> | <b>volatile</b> |
| <b>do</b>       | <b>if</b>     | <b>static</b> | <b>while</b>    |

***Note:** The number of keywords may change depending on the version of C you are using. For example, keywords present in ANSI C are 32 while in C11, it was increased to 44. Moreover, in the latest c23, it is increased to around 54.*

## 2. C Token – Identifiers

Identifiers are used as the general terminology for the naming of variables, functions, and arrays. These are user-defined names consisting of an arbitrarily long sequence of letters and digits with either a letter or the underscore(\_) as a first character. Identifier names must differ in spelling and case from any keywords. You cannot use keywords as identifiers; they are reserved for special use. Once declared, you can use the identifier in later program statements to refer to the associated value. A special identifier called a statement label can be used in goto statements.

### Rules for Naming Identifiers

Certain rules should be followed while naming c identifiers which are as follows:

They must begin with a letter or underscore(\_).

They must consist of only letters, digits, or underscore. No other special character is allowed.

It should not be a keyword.

It must not contain white space.

It should be up to 31 characters long as only the first 31 characters are significant.

***Note:** Identifiers are case-sensitive so names like variable and Variable will be treated as different.*

For example,

**main:** method name.

**a:** variable name.

## 3. C Token – Constants

The constants refer to the variables with fixed values. They are like normal variables but with the difference that their values can not be modified in the program once they are defined.

Constants may belong to any of the data types.

### Examples of Constants in C

```
const int c_var = 20;
```

```
const int* const ptr = &c_var;
```



#### 4. C Token – Strings

Strings are nothing but an array of characters ended with a null character ('\0'). This null character indicates the end of the string. Strings are always enclosed in double quotes. Whereas, a character is enclosed in single quotes in C and C++.

##### Examples of String

“Hello”

“abc@123”

#### 5. C Token – Special Symbols

The following special symbols are used in C having some special meaning and thus, cannot be used for some other purpose. Some of these are listed below:

**Brackets[]:** Opening and closing brackets are used as array element references. These indicate single and multidimensional subscripts.

**Parentheses():** These special symbols are used to indicate function calls and function parameters.

**Braces{}:** These opening and ending curly braces mark the start and end of a block of code containing more than one executable statement.

**Comma (,):** It is used to separate more than one statement like for separating parameters in function calls.

**Colon(:):** It is an operator that essentially invokes something called an initialization list.

**Semicolon(;):** It is known as a statement terminator. It indicates the end of one logical entity. That's why each individual statement must be ended with a semicolon.

**Asterisk (\*):** It is used to create a pointer variable and for the multiplication of variables.

**Assignment operator(=):** It is used to assign values and for logical operation validation.

**Pre-processor (#):** The preprocessor is a macro processor that is used automatically by the compiler to transform your program before actual compilation.

**Period (.):** Used to access members of a structure or union.

**Tilde(~):** Bitwise One's Complement Operator.

#### 6. C Token – Operators

Operators are symbols that trigger an action when applied to C variables and other objects. The data items on which operators act are called operands. Depending on the number of operands that an operator can act upon, operators can be classified as follows:

**Unary Operators:** Those operators that require only a single operand to act upon are known as unary operators. For Example increment and decrement operators.

**Binary Operators:** Those operators that require two operands to act upon are called binary operators. Binary operators can further be classified into:

Arithmetic operators

Relational Operators



|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|          | Logical Operators<br>Assignment Operators<br>Bitwise Operator<br><b>Ternary Operator:</b> The operator that requires three operands to act upon is called the ternary operator. Conditional Operator(?) is also called the ternary operator.<br><b>Special Operators:</b> sizeof(), &, etc..                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |          |
|          | <b>TOPIC:2 Data Types, Constants and Variables</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |
|          | <b>SHORT QUESTIONS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |
| <b>2</b> | <b>What is the correct way to determine the size of a variable? (Mar-2023-OLD)[NLJIET]</b><br><br>The sizeof operator is the most common operator in C. It is a compile-time unary operator and used to compute the size of its operand. It returns the size of a variable. It can be applied to any data type, float type, pointer type variables.<br><br>When sizeof() is used with the data types, it simply returns the amount of memory allocated to that data type. The output can be different on different machines like a 32-bit system can show different output while a 64-bit system can show different of same data types.<br><br><pre> #include &lt;stdio.h&gt;  int main() {  int a = 16;  printf("Size of variable a : %d",sizeof(a));  printf("Size of int data type : %d",sizeof(int));  printf("Size of char data type : %d",sizeof(char));  printf("Size of float data type : %d",sizeof(float));  printf("Size of double data type : %d",sizeof(double));  return 0;  }           </pre> <b>Output</b><br><br>Size of variable a : 4 | <b>1</b> |



|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                 |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
|          | <p>Size of int data type : 4</p> <p>Size of char data type : 1</p> <p>Size of float data type : 4</p> <p>Size of double data type : 8</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                 |
|          | <b>DESCRIPTIVE QUESTIONS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |
| <b>1</b> | <p>Explain primary data types used in C. (Jun-2014-OLD) [NLJIET]</p> <p>Explain basic data types of C. (Jun-2016-OLD) [NLJIET]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p><b>7</b></p> <p><b>3</b></p> |
|          | <p>‘C’ compilers support four fundamental data types. They are as follows –</p> <div data-bbox="316 891 1252 1303" data-label="Diagram"> <pre> graph TD     IntegralType[Integral Type] --&gt; Integer[Integer]     IntegralType --&gt; character[character]     Integer --&gt; Signed[Signed]     Integer --&gt; Unsigned[Unsigned]     Signed --&gt; int[int]     Signed --&gt; shortInt[short int]     Signed --&gt; longInt[long int]     Unsigned --&gt; unsignedInt[unsigned int]     Unsigned --&gt; unsignedShortInt[unsigned short int]     Unsigned --&gt; unsignedLongInt[unsigned long int]     character --&gt; signedChar[signed char]     character --&gt; unsignedChar[unsigned char]     FloatingPointType[Floating point type] --&gt; float[float]     FloatingPointType --&gt; double[double]     FloatingPointType --&gt; longDouble[long double] </pre> </div> <ul style="list-style-type: none"> <li>• Integer</li> <li>• Character</li> <li>• Floating – point</li> <li>• Double precision floating point</li> </ul> <p>Primary Data Types</p> <p>Integral data type</p> <p>Integral data types are used to store whole numbers and characters.</p> <p>It is further classified into two types –</p> <ul style="list-style-type: none"> <li>• Integer data type.</li> <li>• Character data type.</li> </ul> <p>Integer data type</p> <p>This data type is used to store whole numbers.</p> <p>The integer storage are short int, int and long int in both signed and unsigned forms.</p> <p>Character data type</p> <p>Character data type is used to store characters only.</p> |                                 |



| Type                                                                                                                                                                                                                                          | size(in bytes) | Range                     | Control String |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------|----------------|
| short in (or) signed short int                                                                                                                                                                                                                | 1              | -128 to 127               | %h             |
| unsigned short int                                                                                                                                                                                                                            | 1              | 0 to 255                  | %uh            |
| int (or) signed int                                                                                                                                                                                                                           | 4              | -32768 to 32767           | %d or %i       |
| unsigned int                                                                                                                                                                                                                                  | 4              | 0 to 65535                | %u             |
| Long int (or) signed long int                                                                                                                                                                                                                 | 4              | -2147483648 to 2147483647 | %ld            |
| Unsigned long int                                                                                                                                                                                                                             | 4              | 0 to 4294967295           | %lu            |
| <p>These characters are internally stored as integers.<br/>Each character has an equivalent ASCII value.<br/>For example, 'A' has ASCII value 65.</p>                                                                                         |                |                           |                |
| Character Data Types                                                                                                                                                                                                                          |                |                           |                |
| Type                                                                                                                                                                                                                                          | Size(in bytes) | Range                     | Control String |
| Char(or) signed Char                                                                                                                                                                                                                          | 1              | -128 to 127               | %C             |
| unsigned Char                                                                                                                                                                                                                                 | 1              | 0 to 255                  | %c             |
| <p>Floating - point Data type<br/>Floating point data types are used to store real numbers.<br/>'float' is used for 6 digits of accuracy.<br/>'double' is used for 12 digits of accuracy.<br/>More than 12 digits, 'long double' is used.</p> |                |                           |                |



| Floating Point Data Types |                |                              |                |
|---------------------------|----------------|------------------------------|----------------|
| Type                      | size(in bytes) | Range                        | Control String |
| float                     | 4              | 3.4E - 38 to 3.4 E + 38      | %f             |
| double                    | 8              | 1.7 E - 308 to 1.7 E + 308   | %lf            |
| long double               | 16             | 3.4 E - 4932 to 1.1 E + 4932 | %Lf            |

|   |                                                                    |   |
|---|--------------------------------------------------------------------|---|
| 2 | Explain fundamental data types in C language(Aug-2023-OLD)[NLJIET] | 3 |
|---|--------------------------------------------------------------------|---|

‘C’ compilers support four fundamental data types. They are as follows –

Integral Type

|               |                    |                                                                                                                                            |           |  |             |  |               |  |
|---------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-----------|--|-------------|--|---------------|--|
| Integer       |                    | <table><tr><td colspan="2">character</td></tr><tr><td colspan="2">signed char</td></tr><tr><td colspan="2">unsigned char</td></tr></table> | character |  | signed char |  | unsigned char |  |
| character     |                    |                                                                                                                                            |           |  |             |  |               |  |
| signed char   |                    |                                                                                                                                            |           |  |             |  |               |  |
| unsigned char |                    |                                                                                                                                            |           |  |             |  |               |  |
| Signed        | Unsigned           |                                                                                                                                            |           |  |             |  |               |  |
| int           | unsigned int       |                                                                                                                                            |           |  |             |  |               |  |
| short int     | unsigned short int |                                                                                                                                            |           |  |             |  |               |  |
| long int      | unsigned long int  |                                                                                                                                            |           |  |             |  |               |  |

|                     |        |             |
|---------------------|--------|-------------|
| Floating point type |        |             |
| float               | double | long double |

- Integer
- Character
- Floating – point
- Double precision floating point

Primary Data Types

Integral data type

Integral data types are used to store whole numbers and characters.

It is further classified into two types –

- Integer data type.
- Character data type.

Integer data type

This data type is used to store whole numbers.



The integer storage are short int, int and long int in both signed and unsigned forms.

Character data type

Character data type is used to store characters only.

| Type                           | size(in bytes) | Range                     | Control String |
|--------------------------------|----------------|---------------------------|----------------|
| short in (or) signed short int | 1              | -128 to 127               | %h             |
| unsigned short int             | 1              | 0 to 255                  | %uh            |
| int (or) signed int            | 4              | -32768 to 32767           | %d or %i       |
| unsigned int                   | 4              | 0 to 65535                | %u             |
| Long int (or) signed long int  | 4              | -2147483648 to 2147483647 | %ld            |
| Unsigned long int              | 4              | 0 to 4294967295           | %lu            |

These characters are internally stored as integers.

Each character has an equivalent ASCII value.

For example, 'A' has ASCII value 65.

### Character Data Types

| Type                 | Size(in bytes) | Range       | Control String |
|----------------------|----------------|-------------|----------------|
| Char(or) signed Char | 1              | -128 to 127 | %C             |
| unsigned Char        | 1              | 0 to 255    | %c             |

Floating - point Data type

Floating point data types are used to store real numbers.

'float' is used for 6 digits of accuracy.



| <p>‘double’ is used for 12 digits of accuracy.<br/>More than 12 digits, ‘long double’ is used.</p> <table><tr><th colspan="4">Floating Point Data Types</th></tr><tr><th>Type</th><th>size(in bytes)</th><th>Range</th><th>Control String</th></tr><tr><td>float</td><td>4</td><td>3.4E - 38 to 3.4 E + 38</td><td>%f</td></tr><tr><td>double</td><td>8</td><td>1.7 E - 308 to 1.7 E + 308</td><td>%lf</td></tr><tr><td>long double</td><td>16</td><td>3.4 E - 4932 to 1.1 E + 4932</td><td>%Lf</td></tr></table> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                           |                | Floating Point Data Types |   |                                                                                                                                  |                                                                                                                                                                           | Type | size(in bytes)                   | Range                                     | Control String | float                                     | 4                                                                   | 3.4E - 38 to 3.4 E + 38 | %f                                 | double                                            | 8 | 1.7 E - 308 to 1.7 E + 308 | %lf | long double | 16 | 3.4 E - 4932 to 1.1 E + 4932 | %Lf |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------|---|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------------------------------|-------------------------------------------|----------------|-------------------------------------------|---------------------------------------------------------------------|-------------------------|------------------------------------|---------------------------------------------------|---|----------------------------|-----|-------------|----|------------------------------|-----|--|
| Floating Point Data Types                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                           |                |                           |   |                                                                                                                                  |                                                                                                                                                                           |      |                                  |                                           |                |                                           |                                                                     |                         |                                    |                                                   |   |                            |     |             |    |                              |     |  |
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | size(in bytes)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Range                                                                                                                                                                     | Control String |                           |   |                                                                                                                                  |                                                                                                                                                                           |      |                                  |                                           |                |                                           |                                                                     |                         |                                    |                                                   |   |                            |     |             |    |                              |     |  |
| float                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3.4E - 38 to 3.4 E + 38                                                                                                                                                   | %f             |                           |   |                                                                                                                                  |                                                                                                                                                                           |      |                                  |                                           |                |                                           |                                                                     |                         |                                    |                                                   |   |                            |     |             |    |                              |     |  |
| double                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1.7 E - 308 to 1.7 E + 308                                                                                                                                                | %lf            |                           |   |                                                                                                                                  |                                                                                                                                                                           |      |                                  |                                           |                |                                           |                                                                     |                         |                                    |                                                   |   |                            |     |             |    |                              |     |  |
| long double                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3.4 E - 4932 to 1.1 E + 4932                                                                                                                                              | %Lf            |                           |   |                                                                                                                                  |                                                                                                                                                                           |      |                                  |                                           |                |                                           |                                                                     |                         |                                    |                                                   |   |                            |     |             |    |                              |     |  |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | What is difference between keywords and identifiers? Explain rules for naming an identifier. (Jun-2015-OLD) (Jan-2020-OLD) [NLJIET]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                           |                | 4                         |   |                                                                                                                                  |                                                                                                                                                                           |      |                                  |                                           |                |                                           |                                                                     |                         |                                    |                                                   |   |                            |     |             |    |                              |     |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <table><tr><th>SR. NO.</th><th>KEYWORD</th><th>IDENTIFIER</th></tr><tr><td>1</td><td>Keywords are predefined word that gets reserved for working program that have special meaning and cannot get used anywhere else.</td><td>Identifiers are the values used to define different programming items such as variables, integers, structures, unions and others and mostly have an alphabetic character.</td></tr><tr><td>2</td><td>Specify the type/kind of entity.</td><td>Identify the name of a particular entity.</td></tr><tr><td>3</td><td>It always starts with a lowercase letter.</td><td>First character can be a uppercase, lowercase letter or underscore.</td></tr><tr><td>4</td><td>A keyword should be in lower case.</td><td>An identifier can be in upper case or lower case.</td></tr></table> | SR. NO.                                                                                                                                                                   | KEYWORD        | IDENTIFIER                | 1 | Keywords are predefined word that gets reserved for working program that have special meaning and cannot get used anywhere else. | Identifiers are the values used to define different programming items such as variables, integers, structures, unions and others and mostly have an alphabetic character. | 2    | Specify the type/kind of entity. | Identify the name of a particular entity. | 3              | It always starts with a lowercase letter. | First character can be a uppercase, lowercase letter or underscore. | 4                       | A keyword should be in lower case. | An identifier can be in upper case or lower case. |   |                            |     |             |    |                              |     |  |
| SR. NO.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | KEYWORD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | IDENTIFIER                                                                                                                                                                |                |                           |   |                                                                                                                                  |                                                                                                                                                                           |      |                                  |                                           |                |                                           |                                                                     |                         |                                    |                                                   |   |                            |     |             |    |                              |     |  |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Keywords are predefined word that gets reserved for working program that have special meaning and cannot get used anywhere else.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Identifiers are the values used to define different programming items such as variables, integers, structures, unions and others and mostly have an alphabetic character. |                |                           |   |                                                                                                                                  |                                                                                                                                                                           |      |                                  |                                           |                |                                           |                                                                     |                         |                                    |                                                   |   |                            |     |             |    |                              |     |  |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Specify the type/kind of entity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Identify the name of a particular entity.                                                                                                                                 |                |                           |   |                                                                                                                                  |                                                                                                                                                                           |      |                                  |                                           |                |                                           |                                                                     |                         |                                    |                                                   |   |                            |     |             |    |                              |     |  |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | It always starts with a lowercase letter.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | First character can be a uppercase, lowercase letter or underscore.                                                                                                       |                |                           |   |                                                                                                                                  |                                                                                                                                                                           |      |                                  |                                           |                |                                           |                                                                     |                         |                                    |                                                   |   |                            |     |             |    |                              |     |  |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | A keyword should be in lower case.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | An identifier can be in upper case or lower case.                                                                                                                         |                |                           |   |                                                                                                                                  |                                                                                                                                                                           |      |                                  |                                           |                |                                           |                                                                     |                         |                                    |                                                   |   |                            |     |             |    |                              |     |  |



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 5                                                                                                                    | A keyword contains only alphabetical characters.                                  | An identifier can consist of alphabetical characters, digits and underscores.      |   |         |  |        |          |     |              |           |                    |          |                   |           |  |             |               |                     |  |  |       |        |             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---|---------|--|--------|----------|-----|--------------|-----------|--------------------|----------|-------------------|-----------|--|-------------|---------------|---------------------|--|--|-------|--------|-------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 6                                                                                                                    | They help to identify a specific property that exists within a computer language. | They help to locate the name of the entity that gets defined along with a keyword. |   |         |  |        |          |     |              |           |                    |          |                   |           |  |             |               |                     |  |  |       |        |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 7                                                                                                                    | No special symbol, punctuation is used.                                           | No punctuation or special symbol except ‘underscore’ is used.                      |   |         |  |        |          |     |              |           |                    |          |                   |           |  |             |               |                     |  |  |       |        |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 8                                                                                                                    | Examples of keywords are: int, char, if, while, do, class etc.                    | Examples of identifiers are: Test, count1, high_speed, etc.                        |   |         |  |        |          |     |              |           |                    |          |                   |           |  |             |               |                     |  |  |       |        |             |
| <p>The identifiers must conform to the following rules.</p> <ol style="list-style-type: none"><li>1. First character must be an alphabet or underscore.</li><li>2. Identifier names must consist of only letters, digits and underscore.</li><li>3. An identifier name should have less than 31 characters. (32 Characters in Newer version)</li><li>4. Any standard C language keyword cannot be used as a variable name.</li><li>5. An identifier should not contain a space</li></ol>                                                                                                                                                                                                                                 |                                                                                                                      |                                                                                   |                                                                                    |   |         |  |        |          |     |              |           |                    |          |                   |           |  |             |               |                     |  |  |       |        |             |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Describe the four basic data types. How could we extend the range of values they represent? (May-2018-OLD) [NLJIET]  |                                                                                   |                                                                                    | 3 |         |  |        |          |     |              |           |                    |          |                   |           |  |             |               |                     |  |  |       |        |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Describe the four basic data types. How could we extend the range of values they represent? (Jan-2017-OLD [NLJIET]   |                                                                                   |                                                                                    | 4 |         |  |        |          |     |              |           |                    |          |                   |           |  |             |               |                     |  |  |       |        |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Explain four basic data types in C and provision to extend range for appropriate data types. (Nov-2020-NEW) [NLJIET] |                                                                                   |                                                                                    | 4 |         |  |        |          |     |              |           |                    |          |                   |           |  |             |               |                     |  |  |       |        |             |
| <p>‘C’ compilers support four fundamental data types. They are as follows –</p> <div><div><div><div>Integral Type</div><div><table><tr><th colspan="2">Integer</th></tr><tr><th>Signed</th><th>Unsigned</th></tr><tr><td>int</td><td>unsigned int</td></tr><tr><td>short int</td><td>unsigned short int</td></tr><tr><td>long int</td><td>unsigned long int</td></tr></table></div><div><table><tr><th colspan="2">character</th></tr><tr><td>signed char</td><td>unsigned char</td></tr></table></div></div></div><div><table><tr><th colspan="3">Floating point type</th></tr><tr><td>float</td><td>double</td><td>long double</td></tr></table></div></div> <ul style="list-style-type: none"><li>• Integer</li></ul> |                                                                                                                      |                                                                                   |                                                                                    |   | Integer |  | Signed | Unsigned | int | unsigned int | short int | unsigned short int | long int | unsigned long int | character |  | signed char | unsigned char | Floating point type |  |  | float | double | long double |
| Integer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                      |                                                                                   |                                                                                    |   |         |  |        |          |     |              |           |                    |          |                   |           |  |             |               |                     |  |  |       |        |             |
| Signed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Unsigned                                                                                                             |                                                                                   |                                                                                    |   |         |  |        |          |     |              |           |                    |          |                   |           |  |             |               |                     |  |  |       |        |             |
| int                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | unsigned int                                                                                                         |                                                                                   |                                                                                    |   |         |  |        |          |     |              |           |                    |          |                   |           |  |             |               |                     |  |  |       |        |             |
| short int                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | unsigned short int                                                                                                   |                                                                                   |                                                                                    |   |         |  |        |          |     |              |           |                    |          |                   |           |  |             |               |                     |  |  |       |        |             |
| long int                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | unsigned long int                                                                                                    |                                                                                   |                                                                                    |   |         |  |        |          |     |              |           |                    |          |                   |           |  |             |               |                     |  |  |       |        |             |
| character                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                      |                                                                                   |                                                                                    |   |         |  |        |          |     |              |           |                    |          |                   |           |  |             |               |                     |  |  |       |        |             |
| signed char                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | unsigned char                                                                                                        |                                                                                   |                                                                                    |   |         |  |        |          |     |              |           |                    |          |                   |           |  |             |               |                     |  |  |       |        |             |
| Floating point type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                      |                                                                                   |                                                                                    |   |         |  |        |          |     |              |           |                    |          |                   |           |  |             |               |                     |  |  |       |        |             |
| float                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | double                                                                                                               | long double                                                                       |                                                                                    |   |         |  |        |          |     |              |           |                    |          |                   |           |  |             |               |                     |  |  |       |        |             |



- Character
- Floating – point
- Double precision floating point

#### Primary Data Types

#### Integral data type

Integral data types are used to store whole numbers and characters.

It is further classified into two types –

- Integer data type.
- Character data type.

#### Integer data type

This data type is used to store whole numbers.

The integer storage are short int, int and long int in both signed and unsigned forms.

#### Character data type

Character data type is used to store characters only.

| Type                           | size(in bytes) | Range                     | Control String |
|--------------------------------|----------------|---------------------------|----------------|
| short in (or) signed short int | 1              | -128 to 127               | %h             |
| unsigned short int             | 1              | 0 to 255                  | %uh            |
| int (or) signed int            | 4              | -32768 to 32767           | %d or %i       |
| unsigned int                   | 4              | 0 to 65535                | %u             |
| Long int (or) signed long int  | 4              | -2147483648 to 2147483647 | %ld            |
| Unsigned long int              | 4              | 0 to 4294967295           | %lu            |

These characters are internally stored as integers.

Each character has an equivalent ASCII value.

For example, 'A' has ASCII value 65.



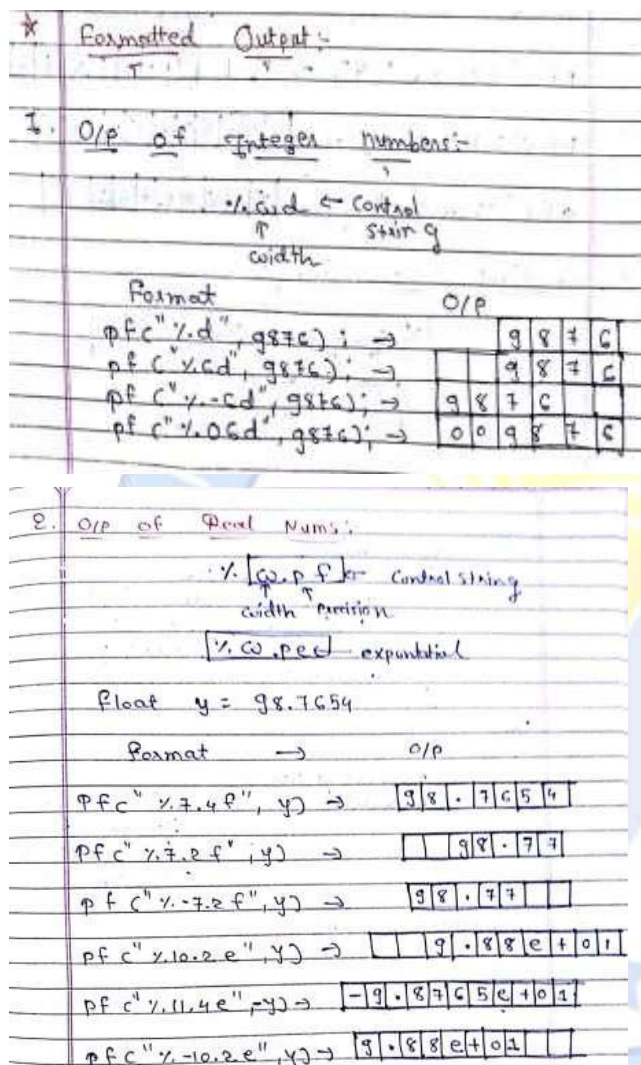
| Character Data Types                                                                                                                                                                                                               |                                                                                                                                                                                                                                                            |                              |                |   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------|---|
| Type                                                                                                                                                                                                                               | Size(in bytes)                                                                                                                                                                                                                                             | Range                        | Control String |   |
| Char(or) signed Char                                                                                                                                                                                                               | 1                                                                                                                                                                                                                                                          | -128 to 127                  | %C             |   |
| unsigned Char                                                                                                                                                                                                                      | 1                                                                                                                                                                                                                                                          | 0 to 255                     | %c             |   |
| Floating - point Data type<br>Floating point data types are used to store real numbers.<br>'float' is used for 6 digits of accuracy.<br>'double' is used for 12 digits of accuracy.<br>More than 12 digits, 'long double' is used. |                                                                                                                                                                                                                                                            |                              |                |   |
| Floating Point Data Types                                                                                                                                                                                                          |                                                                                                                                                                                                                                                            |                              |                |   |
| Type                                                                                                                                                                                                                               | size(in bytes)                                                                                                                                                                                                                                             | Range                        | Control String |   |
| float                                                                                                                                                                                                                              | 4                                                                                                                                                                                                                                                          | 3.4E - 38 to 3.4 E + 38      | %f             |   |
| double                                                                                                                                                                                                                             | 8                                                                                                                                                                                                                                                          | 1.7 E - 308 to 1.7 E + 308   | %lf            |   |
| long double                                                                                                                                                                                                                        | 16                                                                                                                                                                                                                                                         | 3.4 E - 4932 to 1.1 E + 4932 | %Lf            |   |
| 7                                                                                                                                                                                                                                  | Define variable and constant. Explain different types of constants. (May-2018-OLD) [NLJIET]                                                                                                                                                                |                              |                | 4 |
|                                                                                                                                                                                                                                    | Explain different types of constants. (Jan-2020-NEW) [NLJIET]                                                                                                                                                                                              |                              |                | 3 |
|                                                                                                                                                                                                                                    | A <b>variable</b> is a name of the memory location. It is used to store data. Its value can be changed, and it can be reused many times.<br>Syntax :<br>type variable_list;<br>The example of declaring the variable is given below:<br>int a;<br>float b; |                              |                |   |



|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|   | <p>char c;<br/>Here, a, b, c are variables. The int, float, char are the data types.</p> <p><b>A constant</b> is a value or variable that can't be changed in the program, for example: 10, 20, 'a', 3.4, "c programming" etc.</p> <p>List of Constants in C</p> <p>Constant Example</p> <p>Decimal Constant 10, 20, 450 etc.</p> <p>Real or Floating-point Constant 10.3, 20.2, 450.6 etc.</p> <p>Octal Constant 021, 033, 046 etc.</p> <p>Hexadecimal Constant 0x2a, 0x7b, 0xaa etc.</p> <p>Character Constant 'a', 'b', 'x' etc.</p> <p>String Constant "c", "c program" etc.</p> <p>There are two ways to define constant in C programming.</p> <ul style="list-style-type: none"> <li>• const keyword</li> <li>• #define preprocessor</li> </ul> <p>1. C const keyword</p> <pre>#include&lt;stdio.h&gt; int main(){     const float PI=3.14;     printf("The value of PI is: %f",PI);     return 0; }</pre> <p>Output:<br/>The value of PI is: 3.140000</p> <p>2. C #define preprocessor</p> <pre>#include &lt;stdio.h&gt; #define PI 3.14 main() {     printf("%f",PI); }</pre> <p>Output:<br/>3.140000</p> |   |
| 9 | What is formatted output? Using printf() statement explain it. (Jun-2019-NEW) [NLJIET]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3 |
|   | <p>Definition</p> <p>Formatted input output functions enable programmers to manipulate the presentation and</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |



layout of data, whereas unformatted input output functions deal with raw data without any formatting involved.



### TOPIC:3 Operators, Expressions, Evaluation of Expressions, Type Conversion, Precedence and Associativity, I/O Functions

#### SHORT QUESTIONS

- 1** What will be the output of following code? [NLJIET]
- ```
{
    int x = 10, y=15; x = x++;
    y = ++y;
    printf("%d, %d \n", x, y);
}
```

10, 16

**1**



|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 2                     | Justify, 2 and '2' both are not same in C language.(Aug-2022-NEW)[NLJIET]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1 |
|                       | In the C programming language, the number 2 is an integer constant, while the string "2" is a character string constant. These are not the same and cannot be used interchangeably in C. For example, the integer constant 2 can be used in mathematical operations, such as addition and multiplication, while the string "2" cannot. The string "2" must be converted to an integer value before it can be used in such operations.                                                                                                                                                                                                                                                                                                                                                                                       |   |
| 3                     | What is the role of <b>sizeof</b> operator in C language?(Aug-2022-NEW)[NLJIET]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1 |
|                       | The sizeof operator in C is a unary operator that returns the size, in bytes, of a data type or a variable. It can be used to determine the memory requirements of a program, to allocate dynamic memory, and to determine the size of arrays and structures. The sizeof operator is particularly useful when writing code that is platform-independent, as it can be used to determine the size of data types that may vary depending on the hardware architecture.                                                                                                                                                                                                                                                                                                                                                        |   |
| 4                     | Ternary operator can be nested. (True/False) (Aug-2022-NEW)[NLJIET]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1 |
|                       | True. Ternary operator can be nested in C language                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
| 5                     | Explain conditional operator.(Mar-2023-OLD)[NLJIET]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1 |
|                       | <p>The conditional operator is also known as a ternary operator. The conditional statements are the decision-making statements which depends upon the output of the expression. It is represented by two symbols, i.e., '?' and ':'.<br/><br/>As conditional operator works on three operands, so it is also known as the ternary operator.<br/><br/>Syntax of a conditional operator<br/><br/>Expression1? expression2: expression3;</p> <pre>#include &lt;stdio.h&gt; int main() {     int age; // variable declaration     printf("Enter your age");     scanf("%d",&amp;age); // taking user input for age variable     (age&gt;=18)? (printf("eligible for voting")) : (printf("not eligible for voting")); // conditional operator     return 0; }</pre> <p>Output:<br/>Enter your age 20<br/>eligible for voting</p> |   |
| DESCRIPTIVE QUESTIONS |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |



| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Explain various Operators used in C language. (Dec-2013-OLD) (Aug-2023-OLD)[NLJIET]                                  | 7                |          |                  |                |        |                                 |                |               |                      |                 |                      |                      |          |                   |                    |                   |                       |                      |                |                  |    |                                 |                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------|----------|------------------|----------------|--------|---------------------------------|----------------|---------------|----------------------|-----------------|----------------------|----------------------|----------|-------------------|--------------------|-------------------|-----------------------|----------------------|----------------|------------------|----|---------------------------------|------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Discuss about various operators used in C language. (Jan-2015-OLD) (Dec-2015-OLD) [NLJIET]                           | 7                |          |                  |                |        |                                 |                |               |                      |                 |                      |                      |          |                   |                    |                   |                       |                      |                |                  |    |                                 |                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | List out the operators used in C language and explain any three with example. (May-2018-OLD) (Jan-2020-NEW) [NLJIET] | 4                |          |                  |                |        |                                 |                |               |                      |                 |                      |                      |          |                   |                    |                   |                       |                      |                |                  |    |                                 |                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | List all operators used in C language and explain any three with example. (Jan-2019-NEW) [NLJIET]                    | 7                |          |                  |                |        |                                 |                |               |                      |                 |                      |                      |          |                   |                    |                   |                       |                      |                |                  |    |                                 |                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | What is various types of operator available in C. (Jan-2020-OLD) [NLJIET]                                            | 7                |          |                  |                |        |                                 |                |               |                      |                 |                      |                      |          |                   |                    |                   |                       |                      |                |                  |    |                                 |                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Explain different categories of operators available in C programming. (Nov-2020-NEW) [NLJIET]                        | 7                |          |                  |                |        |                                 |                |               |                      |                 |                      |                      |          |                   |                    |                   |                       |                      |                |                  |    |                                 |                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | List out all operators and explain any two in details.(Aug-2023-NEW)[NLJIET]                                         | 4                |          |                  |                |        |                                 |                |               |                      |                 |                      |                      |          |                   |                    |                   |                       |                      |                |                  |    |                                 |                  |
| <table border="1"> <thead> <tr> <th>Operatos</th><th>Type of Operatos</th><th>Operation Type</th></tr> </thead> <tbody> <tr> <td>++, --</td><td>Increments/Decrements Operators</td><td>Unary Operator</td></tr> <tr> <td>+, -, *, /, %</td><td>Arithmetic Operators</td><td rowspan="6">Binary Operator</td></tr> <tr> <td>&lt;, &lt;=, &gt;, &gt;=, ==, !=</td><td>Relational Operators</td></tr> <tr> <td>&amp;&amp;,  , !</td><td>Logical Operators</td></tr> <tr> <td>&amp;,  , &lt;&lt;, &gt;&gt;, ~, ^</td><td>Bitwise Operators</td></tr> <tr> <td>=, +=, -=, *=, /=, %=</td><td>Assignment Operators</td></tr> <tr> <td>sizeof (), &amp; *</td><td>Special Operaots</td></tr> <tr> <td>?:</td><td>Ternary or Conditional Operator</td><td>Ternary Operator</td></tr> </tbody> </table> <p>An operator is a symbol that operates on a value or a variable. For example: <code>+</code> is an operator to perform addition.</p> <p>C has a wide range of operators to perform various operations.</p> <p><b><u>C Arithmetic Operators</u></b></p> <p>An arithmetic operator performs mathematical operations such as addition, subtraction, multiplication, division etc on numerical values (constants and variables).</p> |                                                                                                                      |                  | Operatos | Type of Operatos | Operation Type | ++, -- | Increments/Decrements Operators | Unary Operator | +, -, *, /, % | Arithmetic Operators | Binary Operator | <, <=, >, >=, ==, != | Relational Operators | &&,  , ! | Logical Operators | &,  , <<, >>, ~, ^ | Bitwise Operators | =, +=, -=, *=, /=, %= | Assignment Operators | sizeof (), & * | Special Operaots | ?: | Ternary or Conditional Operator | Ternary Operator |
| Operatos                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Type of Operatos                                                                                                     | Operation Type   |          |                  |                |        |                                 |                |               |                      |                 |                      |                      |          |                   |                    |                   |                       |                      |                |                  |    |                                 |                  |
| ++, --                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Increments/Decrements Operators                                                                                      | Unary Operator   |          |                  |                |        |                                 |                |               |                      |                 |                      |                      |          |                   |                    |                   |                       |                      |                |                  |    |                                 |                  |
| +, -, *, /, %                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Arithmetic Operators                                                                                                 | Binary Operator  |          |                  |                |        |                                 |                |               |                      |                 |                      |                      |          |                   |                    |                   |                       |                      |                |                  |    |                                 |                  |
| <, <=, >, >=, ==, !=                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Relational Operators                                                                                                 |                  |          |                  |                |        |                                 |                |               |                      |                 |                      |                      |          |                   |                    |                   |                       |                      |                |                  |    |                                 |                  |
| &&,  , !                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Logical Operators                                                                                                    |                  |          |                  |                |        |                                 |                |               |                      |                 |                      |                      |          |                   |                    |                   |                       |                      |                |                  |    |                                 |                  |
| &,  , <<, >>, ~, ^                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Bitwise Operators                                                                                                    |                  |          |                  |                |        |                                 |                |               |                      |                 |                      |                      |          |                   |                    |                   |                       |                      |                |                  |    |                                 |                  |
| =, +=, -=, *=, /=, %=                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Assignment Operators                                                                                                 |                  |          |                  |                |        |                                 |                |               |                      |                 |                      |                      |          |                   |                    |                   |                       |                      |                |                  |    |                                 |                  |
| sizeof (), & *                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Special Operaots                                                                                                     |                  |          |                  |                |        |                                 |                |               |                      |                 |                      |                      |          |                   |                    |                   |                       |                      |                |                  |    |                                 |                  |
| ?:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Ternary or Conditional Operator                                                                                      | Ternary Operator |          |                  |                |        |                                 |                |               |                      |                 |                      |                      |          |                   |                    |                   |                       |                      |                |                  |    |                                 |                  |



|   |                                            |
|---|--------------------------------------------|
| - | subtraction or unary minus                 |
| * | multiplication                             |
| / | division                                   |
| % | remainder after division (modulo division) |

### Example 1: Arithmetic Operators

```
// Working of arithmetic operators
#include <stdio.h>
int main()
{
    int a = 9, b = 4, c;

    c = a+b;
    printf("a+b = %d \n", c);
    c = a-b;
    printf("a-b = %d \n", c);
    c = a*b;
    printf("a*b = %d \n", c);
    c = a/b;
    printf("a/b = %d \n", c);
    c = a%b;
    printf("Remainder when a divided by b = %d \n", c);

    return 0;
}
```

### Output

```
a+b = 13
a-b = 5
a*b = 36
a/b = 2
Remainder when a divided by b=1
```

The operators `+`, `-` and `*` computes addition, subtraction, and multiplication respectively as you might have expected.

In normal calculation,  $9/4 = 2.25$ . However, the output is `2` in the program.



It is because both the variables `a` and `b` are integers. Hence, the output is also an integer. The compiler neglects the term after the decimal point and shows answer `2` instead of `2.25`.

The modulo operator `%` computes the remainder. When `a=9` is divided by `b=4`, the remainder is `1`. The `%` operator can only be used with integers.

Suppose `a = 5.0`, `b = 2.0`, `c = 5` and `d = 2`. Then in C programming,

```
// Either one of the operands is a floating-point number
```

```
a/b = 2.5
```

```
a/d = 2.5
```

```
c/b = 2.5
```

```
// Both operands are integers
```

```
c/d = 2
```

### C Increment and Decrement Operators

C programming has two operators increment `++` and decrement `--` to change the value of an operand (constant or variable) by 1.

Increment `++` increases the value by 1 whereas decrement `--` decreases the value by 1. These two operators are unary operators, meaning they only operate on a single operand.

#### **Example 2: Increment and Decrement Operators**

```
// Working of increment and decrement operators
```

```
#include <stdio.h>
int main()
{
    int a = 10, b = 100;
    float c = 10.5, d = 100.5;

    printf("++a = %d \n", ++a);
    printf("--b = %d \n", --b);
    printf("++c = %f \n", ++c);
    printf("--d = %f \n", --d);

    return 0;
}
```



}

**Output**

```

++a = 11
--b = 99
++c = 11.500000
--d = 99.500000

```

Here, the operators ++ and -- are used as prefixes. These two operators can also be used as postfixes like a++ and a--.

**C Assignment Operators**

An assignment operator is used for assigning a value to a variable. The most common assignment operator is =

| Operator | Example | Same as |
|----------|---------|---------|
| =        | a = b   | a = b   |
| +=       | a += b  | a = a+b |
| -=       | a -= b  | a = a-b |
| *=       | a *= b  | a = a*b |
| /=       | a /= b  | a = a/b |
| %=       | a %= b  | a = a%b |

**Example 3: Assignment Operators**

```

// Working of assignment operators
#include <stdio.h>
int main()
{
    int a = 5, c;

    c = a;    // c is 5
    printf("c = %d\n", c);
    c += a;   // c is 10
    printf("c = %d\n", c);
    c -= a;   // c is 5
    printf("c = %d\n", c);
}

```



```
c *= a;    // c is 25
printf("c = %d\n", c);
c /= a;    // c is 5
printf("c = %d\n", c);
c %= a;    // c = 0
printf("c = %d\n", c);

return 0;
}
```

**Output**

```
c = 5
c = 10
c = 5
c = 25
c = 5
c = 0
```

**C Relational Operators**

A relational operator checks the relationship between two operands. If the relation is true, it returns 1; if the relation is false, it returns value 0.

Relational operators are used in decision making and loops.



&gt;=

Greater than or equal  
to5 >= 3 is evaluated to  
1

&lt;=

Less than or equal to

5 <= 3 is evaluated to  
0**Example 4: Relational Operators**

```
// Working of relational operators
#include <stdio.h>
int main()
{
    int a = 5, b = 5, c = 10;

    printf("%d == %d is %d \n", a, b, a == b);
    printf("%d == %d is %d \n", a, c, a == c);
    printf("%d > %d is %d \n", a, b, a > b);
    printf("%d > %d is %d \n", a, c, a > c);
    printf("%d < %d is %d \n", a, b, a < b);
    printf("%d < %d is %d \n", a, c, a < c);
    printf("%d != %d is %d \n", a, b, a != b);
    printf("%d != %d is %d \n", a, c, a != c);
    printf("%d >= %d is %d \n", a, b, a >= b);
    printf("%d >= %d is %d \n", a, c, a >= c);
    printf("%d <= %d is %d \n", a, b, a <= b);
    printf("%d <= %d is %d \n", a, c, a <= c);

    return 0;
}
```

**Output**

```
5 == 5 is 1
5 == 10 is 0
5 > 5 is 0
5 > 10 is 0
5 < 5 is 0
5 < 10 is 1
5 != 5 is 0
5 != 10 is 1
5 >= 5 is 1
5 >= 10 is 0
5 <= 5 is 1
```



5 <= 10 is 1

### C Logical Operators

An expression containing logical operator returns either 0 or 1 depending upon whether expression results true or false. Logical operators are commonly used in decision making in C programming.

Logical  
AND. True  
only if all  
operands  
are true

Logical  
OR. True  
only if  
either one  
operand is  
true

### Example 5: Logical Operators

```
// Working of logical operators

#include <stdio.h>
int main()
{
    int a = 5, b = 5, c = 10, result;

    result = (a == b) && (c > b);
    printf("(a == b) && (c > b) is %d \n", result);

    result = (a == b) && (c < b);
    printf("(a == b) && (c < b) is %d \n", result);
}
```



```

result = (a == b) || (c < b);
printf("(a == b) || (c < b) is %d \n", result);

result = (a != b) || (c < b);
printf("(a != b) || (c < b) is %d \n", result);

result = !(a != b);
printf("!(a != b) is %d \n", result);

result = !(a == b);
printf("!(a == b) is %d \n", result);

return 0;
}

```

### Output

```

(a == b) && (c > b) is 1
(a == b) && (c < b) is 0
(a == b) || (c < b) is 1
(a != b) || (c < b) is 0
!(a != b) is 1
!(a == b) is 0

```

### Explanation of logical operator program

- `(a == b) && (c > 5)` evaluates to 1 because both operands `(a == b)` and `(c > 5)` is 1 (true).
- `(a == b) && (c < b)` evaluates to 0 because operand `(c < b)` is 0 (false).
- `(a == b) || (c < b)` evaluates to 1 because `(a == b)` is 1 (true).
- `(a != b) || (c < b)` evaluates to 0 because both operand `(a != b)` and `(c < b)` are 0 (false).
- `!(a != b)` evaluates to 1 because operand `(a != b)` is 0 (false). Hence, `!(a != b)` is 1 (true).
- `!(a == b)` evaluates to 0 because `(a == b)` is 1 (true). Hence, `!(a == b)` is 0 (false).

### C Bitwise Operators

During computation, mathematical operations like: addition, subtraction, multiplication, division, etc are converted to bit-level which makes processing faster and saves power.

Bitwise operators are used in C programming to perform bit-level operations.



- In C language, the following bit-wise operators are used to perform operations on individual bits of an integer value:
- Bitwise AND (&): This operator performs a bit-wise AND operation on each pair of corresponding bits in the operands. The result is 1 if both bits are 1, and 0 otherwise. For example, consider the expression  $5 \& 3$ . The binary representation of 5 is 0000 0101 and the binary representation of 3 is 0000 0011. The result of the bit-wise AND operation is 0000 0001, which is equal to 1.
- Bitwise OR (|): This operator performs a bit-wise OR operation on each pair of corresponding bits in the operands. The result is 1 if either or both bits are 1, and 0 otherwise. For example, consider the expression  $5 | 3$ . The binary representation of 5 is 0000 0101 and the binary representation of 3 is 0000 0011. The result of the bit-wise OR operation is 0000 0111, which is equal to 7.
- Bitwise XOR (^): This operator performs a bit-wise exclusive OR operation on each pair of corresponding bits in the operands. The result is 1 if exactly one of the bits is 1, and 0 otherwise. For example, consider the expression  $5 \wedge 3$ . The binary representation of 5 is 0000 0101 and the binary representation of 3 is 0000 0011. The result of the bit-wise XOR operation is 0000 0110, which is equal to 6.
- Bitwise NOT (~): This operator inverts the value of each bit in the operand. For example, consider the expression  $\sim 5$ . The binary representation of 5 is 0000 0101. The result of the bit-wise NOT operation is 1111 1010, which is equal to -6 in two's complement representation.



- Left Shift (<<< 1. The binary representation of 5 is 0000 0101. The result of the left shift operation is 0000 1010, which is equal to 10.
- Right Shift (>>): This operator shifts the bits of the first operand to the right by the number of positions specified by the second operand. For example, consider the expression 5 >> 1. The binary representation of 5 is 0000 0101. The result of the right shift operation is 0000 0010, which is equal to 2

### Conditional Operator

The conditional operator is also known as a ternary operator. The conditional statements are the decision-making statements which depends upon the output of the expression. It is represented by two symbols, i.e., '?' and ':'.

As conditional operator works on three operands, so it is also known as the ternary operator.

Syntax of a conditional operator

Expression1? expression2: expression3;

```
#include <stdio.h>
int main()
{
    int age; // variable declaration
    printf("Enter your age");
    scanf("%d",&age); // taking user input for age variable
    (age>=18)? (printf("eligible for voting")) : (printf("not eligible for voting")); // conditional operator
    return 0;
}
```

Output:

Enter your age 20  
eligible for voting

### Special Operators

#### Comma Operator

Comma operators are used to link related expressions together. For example:

```
int a, c = 5, d;
```



### The sizeof operator

The sizeof is a unary operator that returns the size of data (constants, variables, array, structure, etc).

#### Example 6: sizeof Operator

```
#include <stdio.h>
int main()
{
    int a;
    float b;
    double c;
    char d;
    printf("Size of int=%lu bytes\n",sizeof(a));
    printf("Size of float=%lu bytes\n",sizeof(b));
    printf("Size of double=%lu bytes\n",sizeof(c));
    printf("Size of char=%lu byte\n",sizeof(d));

    return 0;
}
```

#### Output

```
Size of int = 4 bytes
Size of float = 4 bytes
Size of double = 8 bytes
Size of char = 1 byte
```

|   |                                                                                                                                                      |   |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 2 | Explain different type of operators used in c language with their precedence and associativity. (Jun-2015-OLD) (Dec-2015-OLD) (Mar-2022-NEW)[NLJIET] | 7 |
|   | Describe precedence and associativity of operators with example. (Jan-2019-NEW) [NLJIET]                                                             | 4 |

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The following tables list the C operator precedence from highest to lowest and the associativity for each of the operators:

| Precedence | Operator | Description                           | Associativity |
|------------|----------|---------------------------------------|---------------|
| 1          | ()       | Parentheses<br>(function call)        | Left-to-Right |
|            | []       | Array Subscript<br>(Square Brackets)  |               |
|            | .        | Dot Operator                          |               |
|            | ->       | Structure Pointer<br>Operator         |               |
|            | ++, —    | Postfix<br>increment,<br>decrement    |               |
| 2          | ++ / —   | Prefix increment,<br>decrement        | Right-to-Left |
|            | + / —    | Unary plus,<br>minus                  |               |
|            | !, ~     | Logical<br>NOT, Bitwise<br>complement |               |



|  |  |   |         |                                                   |               |
|--|--|---|---------|---------------------------------------------------|---------------|
|  |  |   | (type)  | Cast Operator                                     |               |
|  |  |   | *       | Dereference Operator                              |               |
|  |  |   | &       | Addressof Operator                                |               |
|  |  |   | sizeof  | Determine size in bytes                           |               |
|  |  | 3 | *,/,%   | Multiplication, division, modulus                 | Left-to-Right |
|  |  | 4 | +/-     | Addition, subtraction                             | Left-to-Right |
|  |  | 5 | << , >> | Bitwise shift left, Bitwise shift right           | Left-to-Right |
|  |  | 6 | < , <=  | Relational less than, less than or equal to       | Left-to-Right |
|  |  |   | > , >=  | Relational greater than, greater than or equal to |               |



|  |  |    |                    |                                         |               |
|--|--|----|--------------------|-----------------------------------------|---------------|
|  |  | 7  | <b>== , !=</b>     | Relational is equal to, is not equal to | Left-to-Right |
|  |  | 8  | <b>&amp;</b>       | Bitwise AND                             | Left-to-Right |
|  |  | 9  | <b>^</b>           | Bitwise exclusive OR                    | Left-to-Right |
|  |  | 10 | <b> </b>           | Bitwise inclusive OR                    | Left-to-Right |
|  |  | 11 | <b>&amp;&amp;</b>  | Logical AND                             | Left-to-Right |
|  |  | 12 | <b>  </b>          | Logical OR                              | Left-to-Right |
|  |  | 13 | <b>?:</b>          | Ternary conditional                     | Right-to-Left |
|  |  | 14 | <b>=</b>           | Assignment                              | Right-to-Left |
|  |  |    | <b>+= , -=</b>     | Addition, subtraction assignment        |               |
|  |  |    | <b>*= , /=</b>     | Multiplication, division assignment     |               |
|  |  |    | <b>%= , &amp;=</b> | Modulus, bitwise AND assignment         |               |

|    |   |                              |                                            |  |
|----|---|------------------------------|--------------------------------------------|--|
|    |   | $\wedge=,  =$                | Bitwise exclusive, inclusive OR assignment |  |
|    |   | $<<=, >>=$                   | Bitwise shift left, right assignment       |  |
| 15 | , | comma (expression separator) | Left-to-Right                              |  |

Operator Precedence in C

Operator precedence determines which operation is performed first in an expression with more than one operator with different precedence.



**Operator Precedence**

10 + 20 \* 30

Multiply will happen first as \* has *higher* precedence

10 + 600

Now addition will happen as + has *lower* precedence

610

We have got the *solution* for the equation

Operator Associativity

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|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|   | <p>Operator associativity is used when two operators of the same precedence appear in an expression. Associativity can be either from Left to Right or Right to Left.</p> ' data-bbox="169 183 741 365"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |
| 3 | <p>What do you mean by type conversion? Why is it necessary? (Jun-2015-OLD) [NLJIET]</p> <p>What do you mean by type conversion? Give example. (Jan-2019-NEW) [NLJIET]</p> <p>What is type conversion? Briefly explain with suitable example. (Mar-2023-OLD)[NLJIET]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3<br>3<br>3 |
| 5 | <p>Explain type conversion and type casting with example. (Nov-2020-NEW) [NLJIET]</p> <p>Explain type casting with example.(Aug-2022-NEW)[NLJIET]</p> <p>Explain Type casting in C with example(Aug-2023-NEW)[NLJIET]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 4<br>3<br>3 |
|   | <p><b>Type casting</b> is a concept in programming where you change the data type of a variable from one type to another. It's like changing a piece of clay from one shape to another. There are two types of type casting: implicit and explicit.</p>  <ul style="list-style-type: none"> <li>• <b>Implicit Type Casting (Coercion/Type conversion):</b></li> <li>• <b>Explicit Type Casting (Casting):</b></li> </ul> <p><b>Implicit Type Casting:</b></p> <p>In <b>implicit type casting</b>, the programming language automatically converts data from one type to another if needed. For example, if you have an <b>integer</b> variable and you try to assign it to a <b>float</b> variable, the programming language will automatically convert the integer to a float without you having to do anything.</p> <p><b>Implicit Type Casting Example</b></p> <p>Implicit type casting happens automatically by the programming language. It converts a value from one type to another without any explicit instruction from the programmer.</p> <p>// An example of implicit conversion</p> |             |



```
#include <stdio.h>
int main()
{
    int x = 10; // integer x
    char y = 'a'; // character c
    // y implicitly converted to int. ASCII
    // value of 'a' is 97
    x = x + y;
    // x is implicitly converted to float
    float z = x + 1.0;
    printf("x = %d, z = %f", x, z);
    return 0;
}
```

Output

x = 107, z = 108.000000

### Explicit Type Casting:

**Explicit type casting**, also known as **type conversion** or **type coercion**, occurs when the programmer explicitly converts a value from one data type to another. Unlike implicit type casting, explicit type casting requires the programmer to specify the desired data type conversion as (type) expression.

### Explicit Type Casting Examples

Explicit type casting, also known as type conversion, requires a manual instruction from the programmer to convert a value from one type to another. Here are some examples:

#### Floating Point to Integer:

```
#include <stdio.h>
int main()
{
    int x = 5, y=2; // integer x and y
    float z; // float z
    z=(float)x/y; // Explicitly converts int to float
    printf(" %f ",z);
    return 0;
}
```

Output

2.500000



|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 4 | Describe assignment, sizeof( ) and ternary operator with example. (Jun-2019-OLD) [NLJIET]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 7      |
|   | Solution mentioned in Answer of Q.1 above                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |
| 6 | Explain ternary (?:) operator in detail with the example.(Aug-2022-NEW)[NLJIET]<br>Demonstrate the use of ternary operator with an example(Jan-2024-NEW)[NLJIET]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3<br>3 |
|   | <p><b><u>Conditional Operator</u></b></p> <p>The conditional operator is also known as a ternary operator. The conditional statements are the decision-making statements which depends upon the output of the expression. It is represented by two symbols, i.e., '?' and '!'.</p> <p>As conditional operator works on three operands, so it is also known as the ternary operator.</p> <p>Syntax of a conditional operator</p> <p>Expression1? expression2: expression3;</p> <pre>#include &lt;stdio.h&gt; int main() {     int age; // variable declaration     printf("Enter your age");     scanf("%d",&amp;age); // taking user input for age variable     (age&gt;=18)? (printf("eligible for voting")) : (printf("not eligible for voting")); // conditional operator     return 0; }</pre> <p>Output:<br/>Enter your age 20<br/>eligible for voting</p> |        |
| 7 | Explain the working of various bit-wise operators with the example. (Aug-2022-NEW)[NLJIET]<br>Demonstrate the use of bitwise operator with an example (Jan-2024-NEW)[NLJIET]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 7<br>4 |
|   | <p><b>Bitwise AND Operator - &amp;</b></p> <p>The output of bitwise AND is 1 if the corresponding bits of two operands is 1. If either bit of an operand is 0, the result of corresponding bit is evaluated to 0. In C Programming, the bitwise AND operator is denoted by &amp;.</p> <p>Let us suppose the bitwise AND operation of two integers 12 and 25.</p> <p>12 = 00001100 (In Binary)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |



25 = 00011001 (In Binary)

Bit Operation of 12 and 25

00001100

& 00011001

00001000 = 8 (In decimal)

Example 1: Bitwise AND

```
#include <stdio.h>
```

```
int main() {
```

```
    int a = 12, b = 25;
```

```
    printf("Output = %d", a & b);
```

```
    return 0;
```

```
}
```

Output

Output = 8

### Bitwise OR Operator - |

The output of bitwise OR is 1 if at least one corresponding bit of two operands is 1. In C Programming, bitwise OR operator is denoted by |.

12 = 00001100 (In Binary)

25 = 00011001 (In Binary)

Bitwise OR Operation of 12 and 25

00001100

| 00011001

00011101 = 29 (In decimal)

Example 2: Bitwise OR

```
#include <stdio.h>
```

```
int main() {
```

```
    int a = 12, b = 25;
```

```
    printf("Output = %d", a | b);
```

```
    return 0;
```

```
}
```

Run Code

Output

Output = 29

### Bitwise XOR (exclusive OR) Operator - ^

The result of bitwise XOR operator is 1 if the corresponding bits of two operands are opposite. It is denoted by ^.

12 = 00001100 (In Binary)

25 = 00011001 (In Binary)

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Bitwise XOR Operation of 12 and 25

00001100

$\wedge$  00011001

00010101 = 21 (In decimal)

Example 3: Bitwise XOR

```
#include <stdio.h>
```

```
int main() {
```

```
int a = 12, b = 25;
```

```
printf("Output = %d", a ^ b);
```

```
return 0;
```

```
}
```

Output

Output = 21

Shift Operators in C programming

There are two shift operators in C programming:

- Right shift operator
- Left shift operator.

### Right Shift Operator

Right shift operator shifts all bits towards right by certain number of specified bits. It is denoted by  $\gg$ .

212 = 11010100 (In binary)

212  $\gg$  2 = 00110101 (In binary) [Right shift by two bits]

212  $\gg$  7 = 00000001 (In binary)

212  $\gg$  8 = 00000000

212  $\gg$  0 = 11010100 (No Shift)

### Left Shift Operator

Left shift operator shifts all bits towards left by a certain number of specified bits. The bit positions that have been vacated by the left shift operator are filled with 0. The symbol of the left shift operator is  $\ll$ .

212 = 11010100 (In binary)

212  $\ll$  1 = 110101000 (In binary) [Left shift by one bit]

212  $\ll$  0 = 11010100 (Shift by 0)

212  $\ll$  4 = 110101000000 (In binary) = 3392 (In decimal)

Example #5: Shift Operators

```
#include <stdio.h>
```

```
int main() {
```

```
int num=212, i;
```



|                 |                                                                                                                                                                                                                                                                                                                                                    |   |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|                 | <pre> for (i = 0; i &lt;= 2; ++i) { printf("Right shift by %d: %d\n", i, num &gt;&gt; i); } printf("\n"); for (i = 0; i &lt;= 2; ++i) { printf("Left shift by %d: %d\n", i, num &lt;&lt; i);  } return 0; } Right Shift by 0: 212 Right Shift by 1: 106 Right Shift by 2: 53 Left Shift by 0: 212 Left Shift by 1: 424 Left Shift by 2: 848 </pre> |   |
| 8               | Demonstrate the working of bitwise shift and <i>sizeof</i> operator with an example. (Mar-2023-NEW)[NLJIET]                                                                                                                                                                                                                                        | 3 |
|                 | Solution mentioned in Answer of Q.1 and Q.7 Above                                                                                                                                                                                                                                                                                                  |   |
| 9               | Briefly explain <<(shift left) and >>(shift right) operator with suitable example. (Mar-2023-OLD)[NLJIET]                                                                                                                                                                                                                                          | 4 |
|                 | Solution mentioned in Answer of Q.7 Above                                                                                                                                                                                                                                                                                                          |   |
| 10              | <p>Give the output of following C code. (Jan-2024-NEW)[NLJIET]</p> <pre> int main(){ printf("%d",15&lt;2); //ANS 0 printf("%d",15&amp;&amp;2); //ANS 1 printf("%d",7%10); //ANS 7 return0; } O/P: 0 1 7 </pre>                                                                                                                                     | 3 |
|                 |                                                                                                                                                                                                                                                                                                                                                    |   |
| <b>PROGRAMS</b> |                                                                                                                                                                                                                                                                                                                                                    |   |
| 1               | <p>Write a C program to convert Celsius to Fahrenheit and vice versa. (Jun-2016-OLD) (May-2018-OLD)[NLJIET]</p> <pre> /* C Program to convert temperature from Fahrenheit to Celsius and vice versa.*/ #include &lt;stdio.h&gt;  int main() { </pre>                                                                                               | 7 |



|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|   | <pre> float fh,cl; int choice;  printf("\n1: Convert temperature from Fahrenheit to Celsius."); printf("\n2: Convert temperature from Celsius to Fahrenheit."); printf("\nEnter your choice (1, 2): "); scanf("%d",&amp;choice);  if(choice ==1){     printf("\nEnter temperature in Fahrenheit: ");     scanf("%f",&amp;fh);     cl= (fh - 32) / 1.8;     printf("Temperature in Celsius: %.2f",cl); } else if(choice==2){     printf("\nEnter temperature in Celsius: ");     scanf("%f",&amp;cl);     fh= (cl*1.8)+32;     printf("Temperature in Fahrenheit: %.2f",fh); } else{     printf("\nInvalid Choice !!!"); } return 0; } </pre> <p><b>Output:</b></p> <pre> 1: Convert temperature from Fahrenheit to Celsius. 2: Convert temperature from Celsius to Fahrenheit. Enter your choice (1, 2): 1  Enter temperature in Fahrenheit: 98.6 Temperature in Celsius: 37.00 </pre> |   |
| 2 | <p><b>Write a program that reads two numbers from key board and gives their addition, subtraction, multiplication, division and modulo.</b><br/>(Jan-2017- OLD) [NLJIET]</p> <pre> #include &lt;stdio.h&gt; #include&lt;conio.h&gt; int main(){ </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 4 |

```
/* Variable declation */
int firstNumber, secondNumber;
int sum, difference, product, modulo;
float quotient;
printf("Enter First Number: ");
scanf("%d", &firstNumber);
printf("Enter Second Number: ");
scanf("%d", &secondNumber);
/* Adding two numbers */
sum = firstNumber + secondNumber;
/* Subtracting two numbers */
difference = firstNumber - secondNumber;
/* Multiplying two numbers*/

product = firstNumber * secondNumber;
/* Dividing two numbers */

quotient = (float)firstNumber /
secondNumber; /* returns remainder of
after an integer division */ modulo =
firstNumber % secondNumber;
printf("\nSum = %d", sum);
printf("\nDifference = %d", difference);
printf("\nMultiplication = %d", product);
printf("\nDivision = %.3f", quotient);
printf("\nRemainder = %d", modulo);
return 0;
}
```

### Output

**Enter First Number: 25**

**Enter Second Number: 4**

**Sum = 29**



|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|   | <p><b>Difference = 21</b></p> <p><b>Multiplication = 100</b></p> <p><b>Division = 6.250</b></p> <p><b>Remainder = 1</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |
| 3 | <p>Develop an application program to convert and print distance between two cities in meters, feet, inches &amp; centimeters. The distance between two cities (In KM) is input through key board. (Jan-2017-OLD) [NLJIET]</p> <pre> #include&lt;stdio.h&gt; #include&lt;conio.h&gt;  int main() {     float km, m, cm, f, in;      printf("Enter distance in kilometers: ");      scanf("%f", &amp;km);      /* calculate the conversion */      m = km * 1000;      cm = km * 1000 * 100;      f = km * 3280.84;      in = km * 39370.08;      printf("The distance in Feet: %f\n", f);     printf("The distance in Inches: %f\n", in);      printf("The distance in Meters: %f\n", m);      printf("The distance in Centimeters: %f\n", cm); </pre> | 7 |



|   |                                                                                                                                                                                                                                                                                                                                                                                                                          |   |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|   | <pre>return (0);  }</pre> <p><b>Output:</b><br/> Enter distance in kilometers: 2<br/> The distance in Feet: 6561.680176<br/> The distance in Inches: 78740.156250<br/> The distance in Meters: 2000.000000<br/> The distance in Centimeters: 200000.000000</p>                                                                                                                                                           |   |
| 4 | <p>Write C program to find area and perimeter of rectangle. (Jun-2019-OLD) [NLJIET]</p> <pre>#include&lt;stdio.h&gt; #include&lt;conio.h&gt;  int main() { int l = 10, b = 10; int A, P; A = l * b; P = 2 * (l + b); printf("Area of rectangle is : %d", A); printf("\nPerimeter of rectangle is : %d", P); return 0; }</pre> <p><b>Output</b></p> <p>Area of rectangle is : 100<br/> Perimeter of rectangle is : 40</p> | 3 |
| 5 | <p>Write a program to find area of triangle. (Nov-2020-NEW) [NLJIET]</p> <pre>#include&lt;stdio.h&gt; #include&lt;conio.h&gt;  int main() { float base, height, area; printf("Enter the base of the triangle: "); scanf("%f", &amp;base); printf("Enter the height of the triangle: "); scanf("%f", &amp;height);</pre>                                                                                                  | 3 |



|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                   |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
|   | <pre>area = (base * height) / 2; printf("The area of the triangle is: %.2f\n", area); return 0; }</pre> <p><b>Output:</b></p> <p><b>Enter the base of the triangle:1</b><br/> <b>Enter the height of the triangle:1</b><br/> <b>The area of the triangle is:0.50</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                   |
| 6 | <p>Implement a C Program to convert temperature from Fahrenheit to Celsius and vice versa.<br/> <b>(Jan-2020-NEW)[NLJIET]</b><br/> Write a C program to compute Fahrenheit from centigrade (<math>f=1.8*c + 32</math>).<br/> <b>(Aug-2023-NEW)[NLJIET]</b></p> <pre>/* C Program to convert temperature from Fahrenheit to Celsius and vice versa.*/ #include &lt;stdio.h&gt;  int main() {     float fh,cl;     int choice;      printf("\n1: Convert temperature from Fahrenheit to Celsius.");     printf("\n2: Convert temperature from Celsius to Fahrenheit.");     printf("\nEnter your choice (1, 2): ");     scanf("%d",&amp;choice);      if(choice ==1){         printf("\nEnter temperature in Fahrenheit: ");         scanf("%f",&amp;fh);         cl= (fh - 32) / 1.8;         printf("Temperature in Celsius: %.2f",cl);     }     else if(choice==2){         printf("\nEnter temperature in Celsius: ");         scanf("%f",&amp;cl);         fh= (cl*1.8)+32;         printf("Temperature in Fahrenheit: %.2f",fh);     } }</pre> | <p>7</p> <p>7</p> |



|   |                                                                                                                                                                                                                                                                                                                  |   |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
|   | <pre>else{     printf("\nInvalid Choice !!!"); } return 0; }</pre> <p><b>Output:</b></p> <pre>1: Convert temperature from Fahrenheit to Celsius. 2: Convert temperature from Celsius to Fahrenheit. Enter your choice (1, 2): 1  Enter temperature in Fahrenheit: 98.6  Enter temperature in Celsius: 36.4</pre> |   |
| 7 | <p>Write the outputs of the following expression.(Aug-2022-NEW)[NLJIET]</p> <p>i) <math>50 \% 2 / 3 + 2</math> – Ans - 2</p> <p>ii) <math>21 / (\text{int}) 2.5 + 3</math> – Ans - 13</p> <p>iii) <math>(1 &gt; 2) \parallel (2 &lt; 3) \&amp;\&amp; 5 &lt; 1</math> – Ans- 0</p>                                | 3 |

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